

Tommaso Grigoletto

– Postdoctoral fellow –

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Research statement

My research interests lay at the intersection of control system theory, applied mathematics, and physics, with a particular interest in complex systems, probability models and quantum science and technology. My previous work has centered on the development of theoretical and computational methods for modelling, analysing, controlling and reducing quantum dynamical systems.

More broadly, my research aims to establish systematic control-theoretic frameworks for understanding and engineering complex systems, including those arising in emerging quantum technologies.

Research profile at a glance

- Research on model reduction for quantum dynamics and quantum control;
- Author of 9 journal papers (6 as first author) including: IEEE Transactions on Automatic Control, IEEE Transactions on Information Theory, Automatica, Quantum;
- Collaborations with Dartmouth College, Université Paul Sabatier, Inria Paris;
- Teaching experience as main instructor in “Fundamentals of automation engineering”;
- Organizer of the Junior QTech seminar series.

Professional Experience

Research experience

01/07/2025 →30/06/2026	Postdoctoral Fellow University of Padua, Padua, Italy Project: Developing aggregation methods for network model reduction Advisors: Prof. Mattia Zorzi and Prof. Francesco Ticozzi Highlight: Developed an approximation technique for deriving time-dependent reduced generators [15].
01/11/2023 →30/06/2025	Postdoctoral Fellow University of Padua, Padua, Italy Project: Approximate reduction methods for classical and quantum networks Advisor: Prof. Francesco Ticozzi Highlight: Expanded the algebraic model reduction technique toward quantum trajectories [13] and approximate reductions onto slow manifolds [17].
06/08/2022 →10/02/2023	Visiting Scholar Dartmouth College, Hanover, NH (USA) Project: Exact model reduction for continuous-time quantum dynamics Advisor: Prof. Lorenza Viola Highlight: Developed an algebraic model reduction technique for open quantum systems [9].

Teaching experience

Main instructor 2023-2026	Fundamentals of Automation Engineering , Bachelor's degree in Engineering Physics, Ca'Foscari University Three consecutive years, 48 hours each year
Teaching assistant 2022-2023	Control Engineering Laboratory , Master's degree in Control Systems Engineering, University of Padua Two consecutive years, 26 hours each year

Education

01/10/2020 →30/09/2023	Ph.D. in Information Engineering , University of Padua Padua, Italy Thesis: Exact model reduction for quantum systems Advisor: Prof. Francesco Ticozzi Defended March 21st, 2024, with honors (summa cum laude).
29/09/2018 →21/07/2020	M.Sc. in Automation Engineering , University of Padua, Padua, Italy Thesis: Measurement-based switching control for quantum systems Advisor: Prof. Francesco Ticozzi Final grade: 110/110
28/08/2015 →16/07/2018	B.Sc. in Information Engineering , University of Padua, Padua, Italy Thesis: Control system for a rocket soft landing Advisor: Prof. Mauro Bisiacco Final grade: 106/110
Sep. 2010 →Jul. 2015	High school , I.T.I.S. Rossi, Vicenza, Italy Thesis: Control system for a quadcopter Advisor: Prof. Paolo Fumene Ferruglio Final grade: 94/100

Awards, Fellowships and Grants

- 2026 **Seal of Excellence – Marie Skłodowska-Curie Action Postdoctoral Fellowship** (92.4%)
Project: “IMPAQT: Improved Magic state Purity viA Quantum conTrol” with Prof. Alain Sarlette;
- 2022 Gini scholarship, Fondazione Aldo Gini, University of Padua;
- 2020 Ph.D. scholarship, University of Padua;
- 2015 Tekne scholarship, Regione Veneto;
- 2014 Leonardo scholarship, Fondazione Centro Produttività Veneto;
- 2013 Participated to the Robocup World Cup in Eindhoven.

Peer recognition

- 2026 Paper [14] was selected as Editors' Suggestion for Physical Review A.
- 2025 I received the IOP Trusted Reviewer status for an ‘Outstanding’ review report.

Presentations and talks

Conferences

- December 11th 2025 – “**Krylov spaces and algebras for efficient simulation of quantum dynamics**”, QCSC Workshop 2025, Trento Italy;
- September 9th 2025 – “**Exact Model Reduction for Continuous-Time Open Quantum Dynamics**”, Poster presentation at IQIS 2025, Bologna Italy;
- December 2024 – “**Exact Model Reduction for Discrete-Time Conditional Quantum Dynamics**”, Invited session at IEEE Conference on Decision and Control, Milan;
- December 2022 – “**Minimal resources for exact simulation of quantum walks**”, IEEE Conference on Decision and Control, Cancun;

Seminars and Lectures

1. March 12th 2026 – “**Model reduction for complex networks (and quantum systems)**”, Prin kick-off meeting, University of Padua;
2. January 26th 2026 – “**Krylov spaces and algebras for efficient simulation of quantum dynamics**”, QSNOW Workshop 2026, Asiago Italy;
3. March 13th 2025 – “**Exact Model Reduction for Quantum Dynamics**”, Invited talk at Quantum Nano Seminar - Dartmouth College;
4. January 20th 2025 – “**Applications of quantum conditional expectations**”, Two-hour lecture included in the Ph.D. course “Quantum Probability Information and Dynamics” held by Francesco Ticozzi, University of Padua;
5. January 10th 2025 – “**Quantum conditional expectation and the Takesaki Theorem**”, Two-hour lecture included in the Ph.D. course “Quantum Probability Information and Dynamics” held by Francesco Ticozzi, University of Padua;
6. September 22nd 2023 – “**Exact model reduction for Quantum Systems: an algebraic approach**”, Invited talk at Inria Paris, Quantic group;
7. June 27th 2023 – “**Exact model reduction for Quantum Systems: an algebraic approach**”, Seminar talk for the quantum lunch at the department of information engineering, University of Padua;
8. April 2022 – “**Algebraic reduction of Hidden Markov Models**”, Seminar talk at the department of information engineering, University of Padua ;

Student supervision

Bachelor Students

2025	Giacomo Giroto Ca’Foscari	“Experiments on a single qubit quantum computer”, Supervisor: T.G.
2025	Rosanna Mantese Ca’Foscari	“Control of a Boeing v-22 Osprey”, Supervisor: T.G. , Co-Supervisor: Matthias Pezzutto
2025	Marco Grolla Ca’Foscari	“Control of a rocket soft-landing”, Supervisor: T.G. , Co-Supervisor: Matthias Pezzutto
2025	Stefano Santello Ca’Foscari	“Analysis and control of a pendulum actuated with reaction wheels”, Supervisor: T.G. , Co-Supervisor: Matthias Pezzutto
2025	Marco Tesio Ca’Foscari	“Simulation of open quantum systems using quantum computers”, Supervisor: T.G.
2025	Marco Vanzetto Ca’Foscari	“Parameter estimation for closed quantum systems”, Supervisor: T.G.
2024	Giuseppe D’Auria UniPD	“Computer based state observers for magnet systems”, Supervisor: Damiano Varagnolo, Co-Supervisor: T.G.
2024	Jacopo Tomasetig Ca’Foscari	“Modelling and Control of tire-road interaction dynamics”, Supervisor: T.G.
2024	Alessandra Dal Bello Ca’Foscari	“Tracking visual markers for maglev platforms”, Supervisor: T.G. , Co-Supervisor: Damiano Varagnolo

Master Students

Ongoing	Mauro Carpanese UniPD	Supervisor: Francesco Ticozzi, Co-supervisor: T.G.
Ongoing	Francesco Sartori UniPD	Supervisor: Francesco Ticozzi, Co-supervisor: T.G.

International collaborations

- Lorenza Viola, Department of Physics and Astronomy, Dartmouth College;
- Clément Pellegrini, Laboratoire de Statistique et Probabilités, Université Paul Sabatier;
- Alain Sarlette, Quantic group, Inria Paris;
- Thomas Schulte-Herbrüggen, Technical University of Munich;
- Pietro Silvi and Darvin Wanisch from the physics department at the University of Padua.

Professional skills

Reviewing, Service and Outreach

- Associate editor for:
 - ECC 2026;
 - IEEE qCCL 2026.
- Reviewed for journal including:
 - IEEE Control System Letters (11 papers),
 - Automatica (3 papers),
 - International Journal of control (1 paper),
 - Journal of Physics A (1 paper).
- Reviewer for conferences:
 - IEEE Conference on Decision and Control 2021,2022,2024,2025;
 - IEEE American control conference 2025.
- Founder and main organizer of the Junior Qtech seminar cycle: a postdoc led cycle of seminars aimed at merging the skills of physicists, mathematicians, engineers and chemists of the Padua QTech center.
- Co-organizer with Prof. Ticozzi of the Invited session “New tools for estimation, modeling and control of quantum systems” at the Conference on Decision and Control 2024.

Languages

Mother tongue: Italian
Other languages: English ($\simeq$ C1 level)

Programming

Advanced: Python, Matlab, $\text{\LaTeX}$
Experience with: Java, C++

Publications

Legend: *=Shared first-authorship

Journal

- [2] **T. Grigoletto** and F. Ticozzi. “Stabilization via feedback switching for quantum stochastic dynamics”. In: *IEEE Control Systems Letters* 6 (2021), pp. 235–240.
- [4] **T. Grigoletto** and F. Ticozzi. “Algebraic Reduction of Hidden Markov Models”. In: *IEEE Transactions on Automatic Control* 68.12 (2023).
- [5] **T. Grigoletto** and F. Ticozzi. “Model Reduction for Quantum Systems: Discrete-time Quantum Walks and Open Markov Dynamics”. In: *IEEE Transactions on Information Theory* (2025).
- [6] W. Liang, **T. Grigoletto**, and F. Ticozzi. “Dissipative feedback switching for quantum stabilization”. In: *Automatica* 165 (2024), p. 111659.
- [7] **T. Grigoletto** and F. Ticozzi. “Exact Model Reduction for Discrete-Time Conditional Quantum Dynamics”. In: *IEEE Control Systems Letters* (2024).
- [8] M. Cortese, **T. Grigoletto**, F. Ticozzi, and A. Ferrante. “Robust positive model reduction via monotone matrices”. In: *IEEE Transactions on Automatic Control* (2025).
- [9] **T. Grigoletto**, Y. Tao, F. Ticozzi, and L. Viola. “Exact Model Reduction for Continuous-Time Open Quantum Dynamics”. In: *Quantum* 9 (2025), p. 1814.
- [13] **T. Grigoletto**, C. Pellegrini, and F. Ticozzi. “Quantum model reduction for continuous-time quantum filters”. In: *Annales Henri Poincaré* (2025).
- [14] M. Peruzzo, **T. Grigoletto**, and F. Ticozzi. “Reconstructing Quantum States and Expectations via Dynamical Tomography”. In: *Physical Review A* 113 (3 Mar. 2026). Selected as Editors’ Suggestion, p. 032435.

Conference

- [1] T. Benciolini*, T. Grigoletto*, and M. Zorzi. “Image compression by means of the multidimensional circulant covariance extension problem—Revisited”. In: *2020 59th IEEE Conference on Decision and Control (CDC)*. IEEE. 2020, pp. 280–285.
- [3] **T. Grigoletto** and F. Ticozzi. “Minimal resources for exact simulation of quantum walks”. In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. 2022, pp. 5155–5160.
- [11] M. Peruzzo, **T. Grigoletto**, and F. Ticozzi. “Reconstructing Quantum States from Local Observation: A Dynamical Viewpoint”. In: *2024 IEEE 63rd Conference on Decision and Control (CDC)*. 2024.
- [16] **T. Grigoletto**, L. Viola, and F. Ticozzi. “Model reduction for controlled quantum Markov dynamics”. In: *European Control Conference (ECC) 2026*. Accepted, yet to appear. 2025.

Under review

- [15] **T. Grigoletto**. *Computing time-dependent reduced models for classical and quantum dynamics*. Submitted to Automatica. 2025.
- [17] **T. Grigoletto**, A. Sarlette, F. Ticozzi, and L. Viola. *Approximate Reduced Lindblad Dynamics via Algebraic and Adiabatic Methods*. Submitted to Quantum Science and Technology. 2026.

Other references

- [10] **T. Grigoletto**. “Exact model reduction for quantum systems”. Ph.D. dissertation. University of Padua, 2024.
- [12] **T. Grigoletto**, A. Gallina, E. Bellesso, and G. Pillonetto. *Applied Functional Analysis and Machine Learning*. Lecture notes. 2024.

In preparation

- [18] **T. Grigoletto**, M. Casanova, L. Viola, and F. Ticozzi. *Weak symmetries and where to find them*. In preparation.
- [19] **T. Grigoletto** and A. Sarlette. *On the well posedness of adiabatic elimination at higher orders*. In preparation.
- [20] **T. Grigoletto**, T. Benoist, C. Pellegrini, L. Viola, and F. Ticozzi. *Symmetries for quantum trajectories*. In preparation.